Top 1000 Viewed Youtube Videos

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Our Dataset

The 1000 Most-Watched YouTube Videos

(https://www.kaggle.com/samithsachidanandan)

Most popular 1000 Youtube videos.csv (89.71 kB)

Detail Compact Column 7 of 7 columns

# rank Video's position	Video Title/name.	# Video views Total views.	# Likes Positive reactions.	# Dislikes Negative reactions.	Category Content type.
1 1000	997 unique values	44.9k 6.55b	433 44.4m	0 178k	Music 43% People & Blogs 26% Other (312) 31%
1	Lil Nas X - Old Town Road (Official Movie) ft. Billy Ray Cyrus	54,071,677	3,497,955	78,799	Music
2	20 Tennis shots if they were not filmed, NOBODY would believe them	3,471,237	19,023	859	
3	JoJo Siwa - Karma (Official Video)	34,206,747	293,563		Music
4	David Kushner - Daylight (Official Music Video)	18,558,390	680,732		Music
5	Wiz Khalifa - See You Again ft. Charlie Puth [Official Video]	6,547,981,039	44,428,537		Music

Lil Nas X - Old Town Road (Official Movie) ft. Billy Ray Cyrus

727M views • 6 years ago

Lil Nas X

Official video for Lil Nas X's Billboard #1 hit, "**Old Town Road** (Remix)" featuring Billy Ray Cyrus. Special guest appearances from ...

CC

20 Tennis Shots If They Weren't Filmed NOBODY Would Believe Them!

8.8K views • 2 years ago

Raz Ols

Tennis most amazing and ridiculous **shots** and points! Federer, Alcaraz, Nadal and many more! Support my work so **I can** invest ...

JoJo Siwa - Karma (Official Video)

54M views • 1 year ago

JoJo Siwa

Official Video for "**Karma**" by **JoJo Siwa** Listen to & Download "**Karma**" out now: https://**jojosiwa**.lnk.to/**Karma** Amazon Music: ...

4K

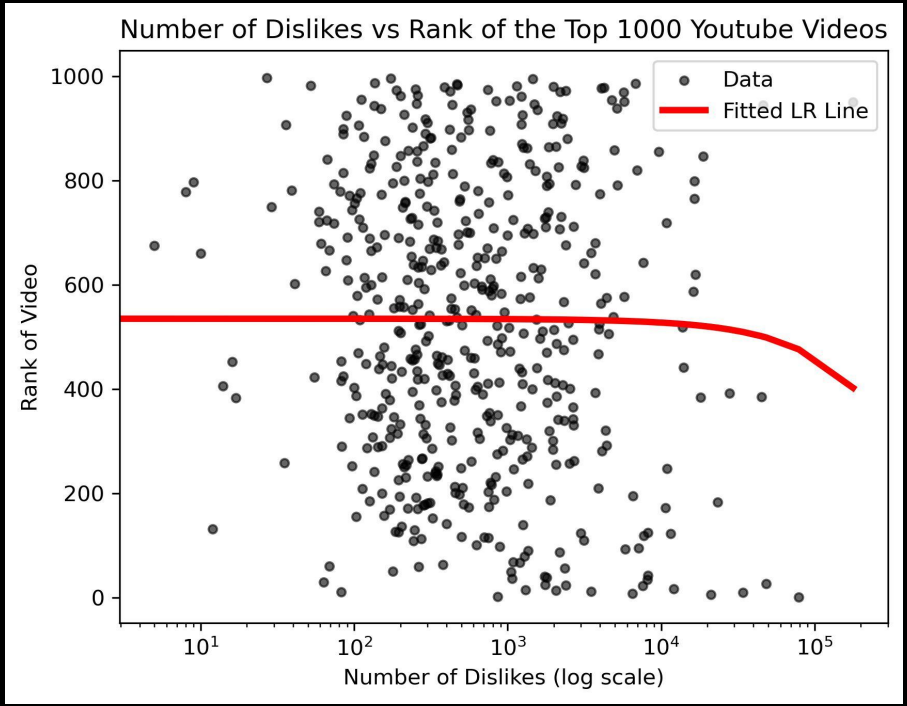
Target Audience

Youtube Influencers looking to grow their outreach with respect to rank.

Is rank a meaningful metric for content creators to consider when planning videos?

Data Cleaning

- Blank Categories hard coded as "Not Specified"
- Removal of Dislikes column
- Log-scale
- Ignore Category variable in our models



Models

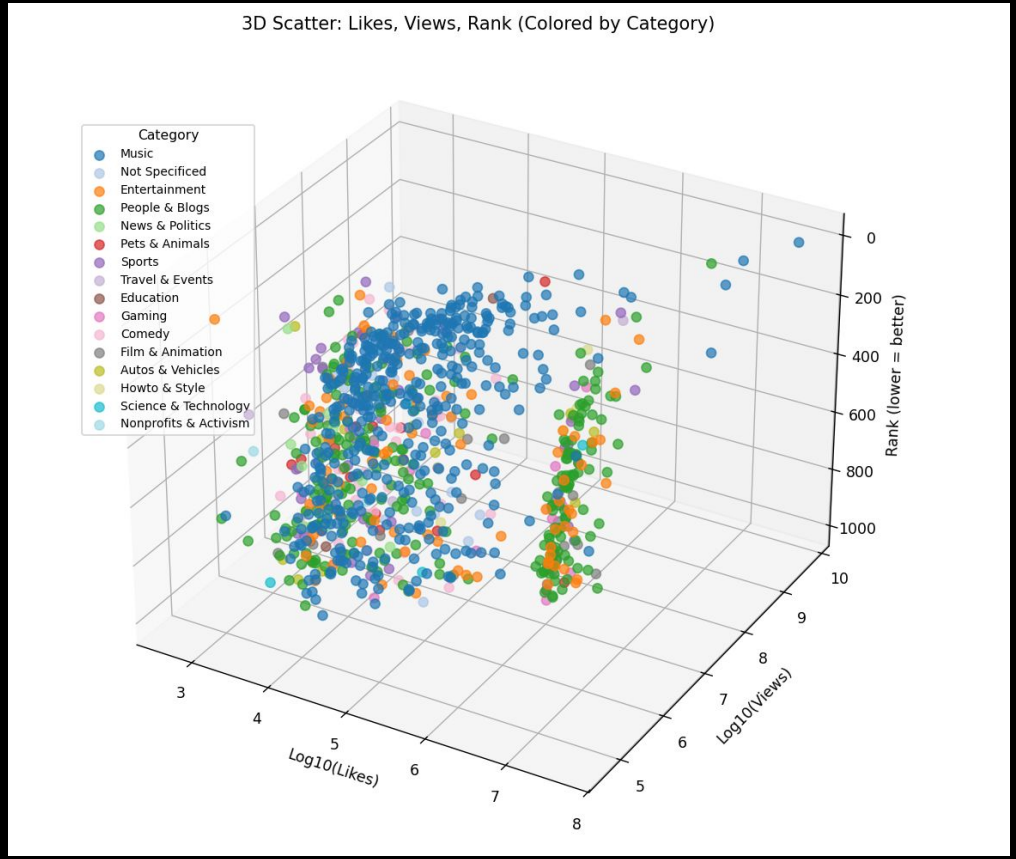
X = Categories, Likes, Views, Published
y = Rank

- Multiple Linear Regression
- kNN
- RandomForest
- PCA

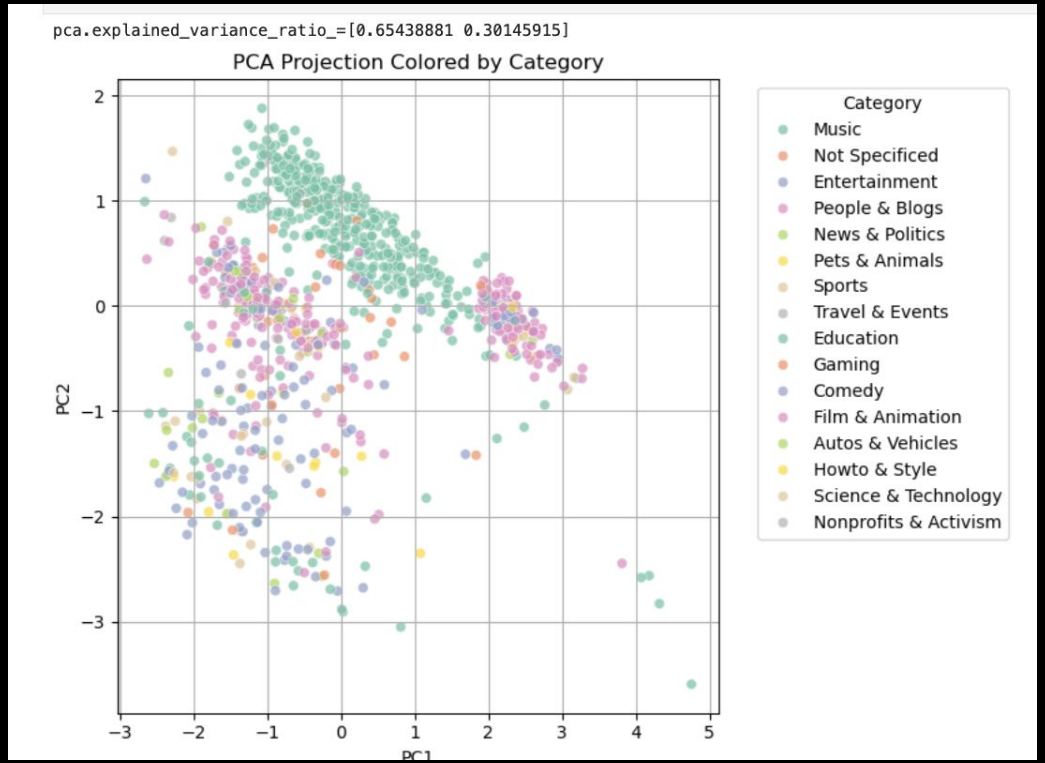
kNN

```
True rank: 812
Predicted rank: 493.6
Error: 318.4
```

	rank	Video views	Likes
rank	1.000000	-0.083281	-0.051359



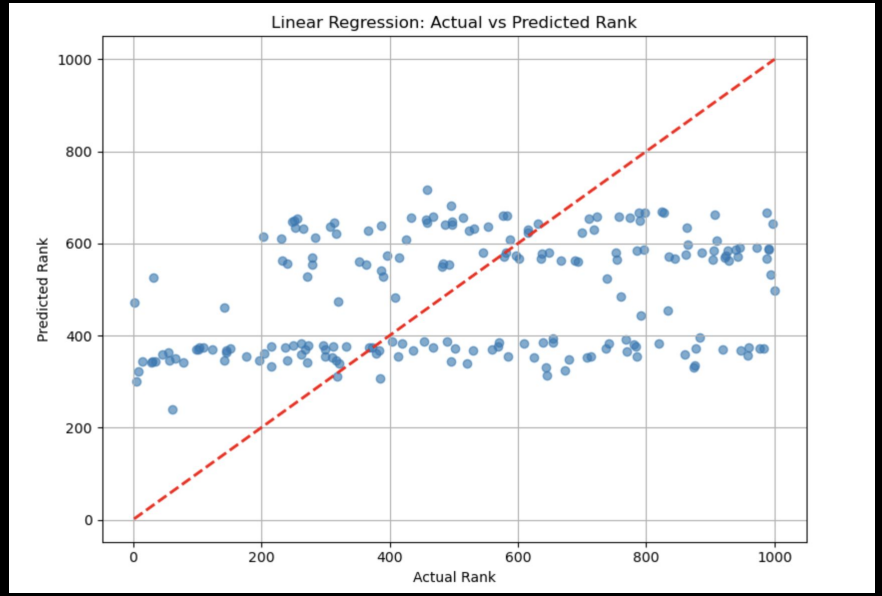
PCA

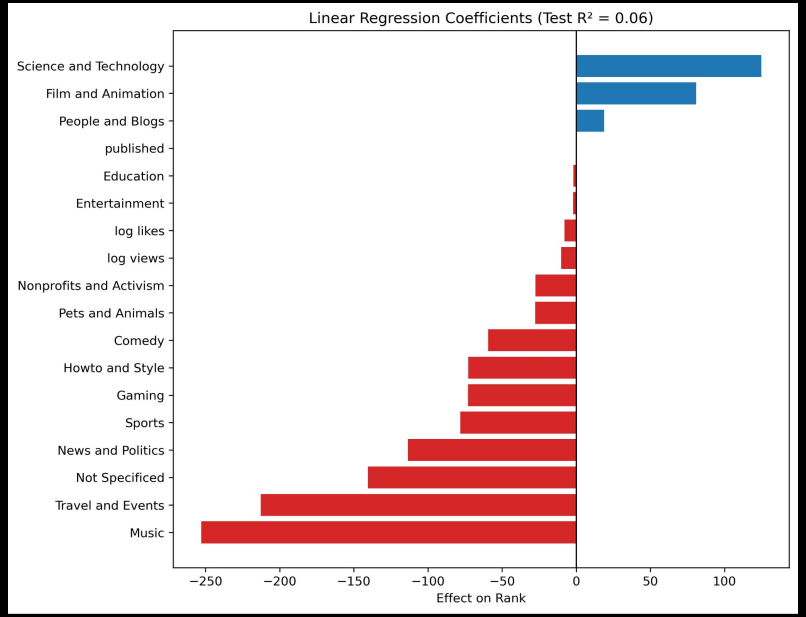



```
pca.explained_variance_ratio_=[0.65438881 0.30145915]
```

- Good Model Fit
- Non-Predictive Model

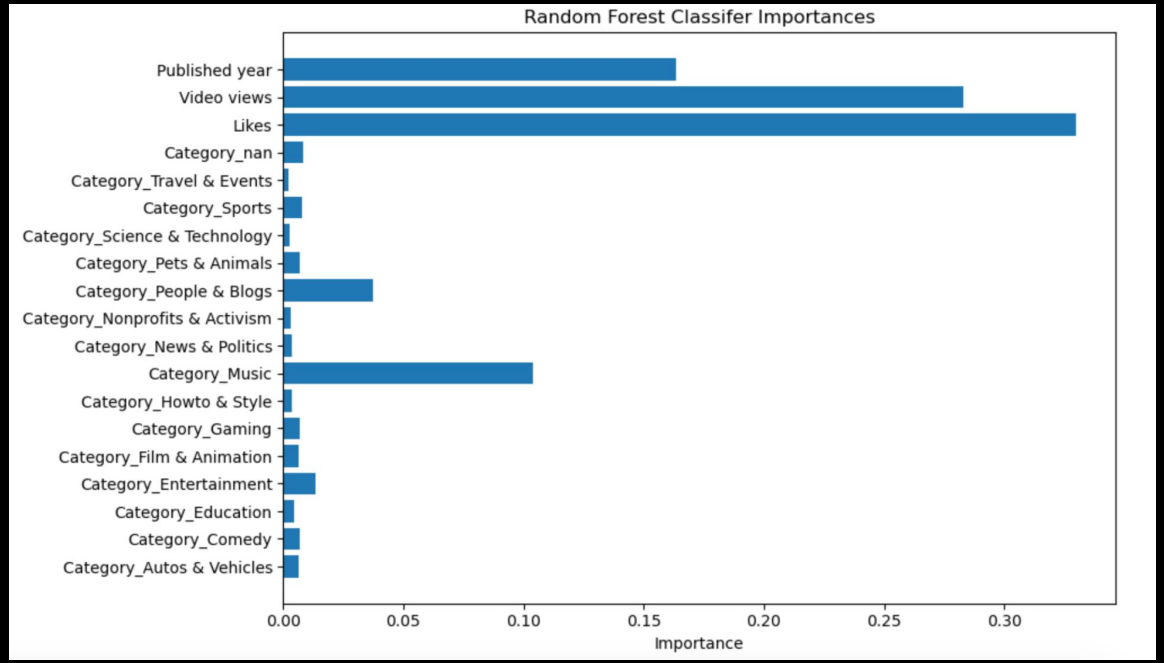
Linear Regression





Random Forest

Validation score: 0.53



Rank is unpredictable.

Challenges

- Dropped a predictor → ended up being unreliable
- Virality is very unpredictable!
- Several other variables that actually matter for the rank according to Youtube- ex: watch time and audience retention- variables not included in this dataset.

Future Considerations

- Other Features to Observe
  - Using subscriber count as a predictor
  - Appearance of the thumbnail
  - How many times the video is clicked
  - Audience watch time
- Research on Rank Factors
- Other Model Strategies

Questions?